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Frontline Staff and Trainer Perspectives on Implementing and Adhering to Positive Behaviour Support in Intellectual Disabilities Care: A Mixed-Methods Study

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ABSTRACT

Background: We aimed to identify key factors in implementing and adhering to positive behaviour support (PBS), provided by frontline staff, targeting challenging behaviours of people with intellectual disabilities living in residential group homes.

Method: Using semi-structured interviews with 12 frontline staff members, we collected perspectives on a PBS training that they received. In two separate focus groups with nine frontline staff members and nine PBS trainers, we discussed factors perceived to influence PBS implementation and adherence. Hybrid coding and thematic analysis were employed.

Results: Frontline staff were positive about the content of the training and noticed improvements in challenging behaviours of the people they cared for. Key factors for implementation and adherence included organisation-wide embedding of PBS, management support, motivating staff, clear documentation, regular evaluations, and additional training sessions.

Conclusions: This study provides valuable insights for the implementation of and adherence to frontline staff-provided PBS.

1 | Introduction

Positive behaviour support (PBS) provided by frontline staff is an effective intervention to decrease challenging behaviours (Konstantinidou et al. 2023; MacDonald and McGill 2013; Grey and McClean 2007) and increase the quality of life of people with intellectual disabilities (Bruinsma et al. 2024; Klaver et al. 2020). Frontline staff provide day-to-day care and emotional support to people with intellectual disabilities and challenging behaviours living in residential facilities. They assist them in daily life activities such as eating, personal hygiene, and leisure activities. Previous research has shown that frontline staff have a great influence on the maintenance or reduction of challenging behaviours (Cox et al. 2015; Simons et al. 2020). Therefore, teaching frontline staff to apply techniques to reduce such challenging behaviours is important, and PBS provides a structured, evidence-based method to reach this.

According to a recent definition of Gore et al. (2022), PBS is a comprehensive, evidence-based framework aimed at enhancing quality of life for people with intellectual disabilities. PBS integrates a biopsychosocial model, emphasising the understanding of behaviours within their biological, psychological, and social contexts. PBS employs a multi-tiered approach, incorporating person-centred planning and proactive interventions. A key feature of PBS is the principle of never employing aversive, restrictive, and abusive practices. PBS emphasises the importance of the rights and values of people with intellectual disabilities and challenging behaviours, to promote self-determination and social inclusion. In addition to its positive effects on the quality of life of people with intellectual disabilities (Bruinsma et al. 2024), frontline staff-provided PBS has positive effects on staff confidence (Lowe et al. 2007), skills, knowledge, attributions, and emotional responses regarding challenging behaviours (MacDonald and McGill 2013). Also, evidence suggests that the delivery of PBS by frontline staff may be cost-effective and reduces the burden on informal care (i.e., family members or close others; Hunter et al. 2020). Yet, implementing (i.e., putting something into practice) and adhering to (i.e., following guidelines or protocols) frontline staff-provided PBS in daily practice appears to be challenging (Bosco et al. 2019; Brady et al. 2024; Campbell 2007; Hassiotis et al. 2018). The scarce literature on effective PBS implementation states the necessity of comprehensive knowledge about PBS principles among all stakeholders, policy alignment, and the crucial role of informing about the benefits and positive outcomes of PBS to increase engagement among stakeholders (Hayward et al. 2021). Regarding successful adherence, regular feedback, staff involvement (Brady et al. 2019), management support (Konstantinidou et al. 2023), and structured evaluation (McKenzie et al. 2020) are deemed essential.

Several factors have been identified that complicate the implementation of and adherence to frontline staff-provided PBS in daily practice. Regarding implementation, there is a lack of clear PBS manuals (Hassiotis et al. 2018; Macdonald et al. 2018), and frontline staff members may have prejudices against behavioural interventions such as PBS, based on, for example, the mistaken belief that these interventions are primarily 'rule-governed' and focus on aversive techniques (Ager and O'May 2001; Lydon et al. 2015). Concerning the adherence to staff-provided PBS, frontline staff may find it difficult to understand the rationale of PBS (Ager and O'May 2001). Moreover, PBS support plans are often written by behavioural specialists with master's degrees, using vocabulary that does not align with the educational level (e.g., vocational education) of the frontline staff members, who have to implement these support plans. This mismatch can result in frontline staff not fully understanding the plans they are supposed to implement (Adkins et al. 2002; Singh et al. 2009; Wardale et al. 2018). Furthermore, PBS is labour-intensive and frontline staff members already experience a lack of time to provide sufficient care (Lowe et al. 2007). Also, high frontline staff turnover is common, making it difficult to consistently provide PBS (Bosco et al. 2019).

In our recent trial on the effectiveness of frontline staff-provided PBS, most of these factors were taken into account by using clear PBS manuals that fitted the educational level of staff (Bruinsma et al. 2024). Frontline staff members received intensive training based on sessions and assignments that were adjusted to the needs and available time of staff. The study showed that frontline staff-provided PBS reduced challenging behaviours of people who exhibited higher levels of challenging behaviours and of people with severe or profound levels of intellectual disabilities. Moreover, the quality of life of all people improved (Bruinsma et al. 2024).

To gain more insight into factors that may contribute to implementing and adhering to frontline staff-provided PBS, the current mixed-method study explored frontline staff perspectives on the PBS training they received, as well as frontline staff and PBS trainers' perspectives on key factors that promote successful implementation and adherence to frontline staff-provided PBS. With this study, we complement our recent trial (Bruinsma et al. 2024) in which we quantitatively examined the effectiveness of frontline staff-provided PBS.

2 | Method

2.1 | Study Design

We conducted a mixed-methods study with semi-structured interviews and focus groups. In the first phase, we conducted semi-structured interviews regarding frontline staff's perspectives on a PBS staff training that was previously examined for effectiveness in our recent trial (Bruinsma et al. 2024). Phase two consisted of one focus group with frontline staff and one with trainers to identify factors influencing the implementation of and adherence to frontline staff-provided PBS.

2.2 | Ethical Approval

The Medical Ethical Committee of the University Centre Groningen determined that our study did not fall under the scope of the Medical Research Involving Human Subjects Act (WMO). The study was conducted in full compliance with the World Medical Association (WMA) Declaration of Helsinki. Data were collected and stored in accordance with the General Data Protection Regulation (GDPR, in Dutch: 'Algemene Verordering Gegevensbescherming'). Informed consent was obtained from all participants as part of this process.

2.3 | Participants

In the first phase of the study, participants were frontline staff members who had received a PBS staff training in our recent trial (Bruinsma et al. 2024). One-hundred-and-twenty-nine frontline staff members from 14 residential group homes were eligible for the current study. Our aimed sample size for both the semi-structured interviews as well as the frontline staff focus group was 12 participants, which is considered sufficient to achieve saturation as well as case-oriented analysis in qualitative research (Hennink and Kaiser 2022; Sandelowski 1995). If more than 12 frontline staff members were willing to participate in the study, random selection was conducted to exclude frontline staff members from the group home with the highest number of applicants. This way, we aimed to enhance the variety of perspectives represented in our study.

In the second phase of the study, frontline staff members participating in the first phase as well as 14 trainers of the PBS staff training in the trial were eligible. Of these 14 trainers, two were the psychologists of group homes included in the trial (i.e., professionals responsible for the diagnosis and support plans of people with intellectual disabilities; see Bruinsma et al. 2024). Twelve trainers were (applied) behavioural therapists who were not affiliated with the group home receiving the training.

2.4 | Procedures

We contacted all 129 frontline staff members who had participated in the training condition of the trial (Bruinsma et al. 2024) by email with background information about the study and a request to participate in phase one (interviews). We specifically stated that participation was voluntary and independent of the previous trial. Moreover, we informed frontline staff about the financial compensation they would receive if they participated in the study (i.e., € 75, per study phase). After informed consent, they were interviewed in January and February 2020.

For the second phase of the study, frontline staff members who participated in phase 1 of the study were contacted by email to plan a date for the focus group. This focus group took place on April 20, 2020. We had to conduct this in an online Zoom meeting due to quarantine restrictions due to the Covid-19 pandemic. The meeting lasted 90 min and was led by a mediator (BJvdH). A research assistant was present for assistance with possible technical difficulties.

Recruitment of trainers for the focus groups was executed in a similar way. All 14 trainers who had provided a PBS training for frontline staff were contacted by email. Trainers were informed about the rationale of the study and the financial compensation (i.e., €150) for participation. The trainers focus group was organised on April 24, 2020, and also had to take place in an online Zoom meeting of 90 min. The focus group was led by a mediator (AdB) and again a research assistant was present.

Besides the data collected as described above, we used data from the trial regarding frontline staff's age, sex, educational level, work experience and willingness to apply new interventions (Bruinsma et al. 2024).

2.5 | PBS Training

The PBS training included eight bi-weekly sessions of 180 min each. The training sessions were led by two trainers with extensive practical and theoretical experience in the use of PBS techniques who were either licensed behavioural therapists or had completed at least a foundational course in cognitive behavioural therapy. All trainers received a one-day workshop on delivering this specific manualised PBS training, followed by five supervision sessions with the PBS training developers during the first team training they delivered. The training consisted of four phases: (1) acquiring observational skills, applying functional behaviour analyses, clearly defining the targeted (i.e., challenging) behaviour to be addressed, and formulating appropriate (i.e., desired) behaviour goals. This initial phase set the groundwork for identifying and understanding the behaviours requiring intervention; (2) implementing antecedent interventions, such as environmental adjustments and personal support aimed at reducing the probability of the targeted behaviour. These interventions were designed to modify the conditions that precede the challenging behaviour, thus preventing its occurrence; (3) managing consequences by reinforcing desired behaviours and teaching skills to replace the challenging behaviour. This phase focused on enhancing the individual's ability to engage in appropriate behaviours as alternatives to the challenging behaviours and to reinforce these; and (4) strengthening and generalising the learned staff skills to other contexts and behaviours. This final phase ensured that staff could apply the techniques across various situations and to different behaviours, enhancing the overall effectiveness of the intervention.Frontline staff members, who provided daily care to individuals with intellectual disabilities and challenging behaviours, actively participated by completing homework assignments like recording interactions or filling out observation schedules. These assignments were reviewed in subsequent sessions, enhancing practical application and adherence to the PBS framework. The training materials included detailed manuals tailored to the educational level of the frontline staff, explaining the PBS rationale and providing tools for effective practice. More information regarding the PBS training is provided in Bruinsma et al. (2024).

2.6 | Measures

To measure frontline staff's perspectives on the PBS training, we conducted semi-structured interviews for which we used the Measurement Instrument for Determinants of Innovations-Intellectual Disabilities (MIDI-ID; Steenbergen et al. 2019). The MIDI-ID is an adjusted version of the Measurement Instrument for Determinants of Innovations (MIDI; Fleuren et al. 2014). It specifically measures determinants that may affect the implementation of health care innovations for people with intellectual disabilities, in order to better target innovation strategies. The MIDI-ID measures four categories of innovation determinants: (1) characteristics of the innovation itself (i.e., PBS training). An example of an item is: 'The innovation is based on the right knowledge'; (2) the user (i.e., frontline staff member). An example of an item is: 'To which extent does the innovation offer personal advantages?'; (3) the organisation, an example of an item is: 'Are sufficient financial means available to conduct the innovation as meant?'; and (4) the socio-political environment. An example of an item is: 'The innovation follows current laws and regulations'. An overview of the four categories and all determinants measured by the MIDI-ID is presented in Table 1. To date, there are no studies on the psychometric properties of the MIDI or the MIDI-ID.

For both focus groups, we created scripts and topic lists in the following order: (1) a summary of the results of the semi-structured interviews was presented (i.e., the perspectives of frontline staff with the PBS training); (2) participants discussed their perspectives on key factors for the implementation of frontline staff-provided PBS (divided into requirements for frontline staff, psychologists, management, and organisation); and (3) participants discussed their perspectives on key factors for adherence to frontline staff-provided PBS and what these imply for different levels within an organisation (i.e., frontline staff, psychologists, management, and organisation). For example, at the management level, a factor could be providing time to apply PBS, or at the organisational level, a factor could be that PBS should be in line with the current developments within the organisation.

The Importance subscale of the Readiness, Efficacy, Attributions, Defensiveness, and Importance Scale-Short Form (READI-SF;

TABLE 1 | Overview of all determinants measured by the MIDI-ID specified by category (Steenbergen et al. 2019).

Determinants regarding the innovation itself
Procedural clarity
Correctness
Completeness
Complexity
Compatibility
Observability
Relevance for client
Determinants regarding the user
Personal benefits/drawbacks
Outcome expectations
Job description
Client satisfaction
Client cooperation
Social support
Descriptive norm
Subjective norm
Self-efficacy
Knowledge
Awareness of content of innovation
Determinants regarding the organisation
Formal ratification by management
Replacement when staff leave
Staff capacity
Financial resources
Time available
Material resources and facilities
Coordinator
Unrest in organisation
Information accessible about use of innovation
Feedback to user about innovation process
Determinants regarding socio-political context
Legislation and regulation

Proctor et al. 2018) was conducted at baseline in the main trial (Bruinsma et al. 2024) to examine staff's willingness to apply new interventions. This subscale assesses caregivers' conviction regarding the significance of the intervention. As the READI-SF is originally a parent instrument, the subscale Importance evaluates parents' perceived importance of interventions and consists of five items that are rated on a 5-point Likert scale, ranging from 'strongly disagree' (1) to 'strongly agree' (5). For our study, we modified the items from parent-perceived importance to staff-perceived importance (Bruinsma et al. 2024).

2.7 | Statistical Analyses

First we checked the representativeness of the current frontline staff sample in comparison with the larger sample of the main trial (Bruinsma et al. 2024) on age, sex, educational level, work experience in years, and willingness to apply new interventions. We conducted multilevel analyses with three levels: organisation, group home, and individual frontline staff member using data from the trial, demographic variables (i.e., age, sex, educational level, work experience in years) and the READI-SF scores.

The answers in the interviews and focus groups were transcribed verbatim. After that, ATLAS.ti (2002–2019) was used for coding the transcripts (according to subject discussed) and analysing the data. We employed a thematic analysis approach which involves identifying, analysing, and reporting patterns (themes) within data (Braun and Clarke 2006). This approach was supplemented by a hybrid coding strategy, meaning we mixed predetermined codes with open coding (Fereday and Muir-Cochrane 2006). This method allowed us to maintain flexibility in data analysis while ensuring alignment with established theoretical constructs, effectively capturing the nuanced perspectives of the participants. The predetermined codes for the semi-structured interviews were based on the four categories of the MIDI-ID. The predetermined codes for the focus groups were the topics of the focus group scripts (i.e., the critical determinants for adherence to frontline staff-provided PBS discussed at frontline staff level, psychologists level, management level and the organisation as a whole, and determinants for implementation of PBS, on frontline staff level, psychologists level, management level and the whole organisation). Moreover, open codes were defined without prior assumptions or categories and were determined based on the content of the transcripts (i.e., transcripts of the semi-structured interviews as well as the focus groups). Multiple codes (i.e., predetermined and open codes) could apply to a single quote.

The transcripts of the semi-structured interviews were coded in a three-step process by a master's student and checked by the first author (E.B.) and the student's supervisor. First, initial open coding was applied, and quotes were categorised by the predetermined codes whilst quotes could still be traced to the participant. Second, open codes were checked for similarity and distinctiveness, merging and combining open codes with similar meaning. Third, patterns within categories were distinguished and analysed. The focus groups were coded independently by two members of the research team (E.B., first author and L.A., research assistant). Coding was conducted in a similar three-step process as the semi-structured interviews. It started with categorisation of the transcript due to the predetermined codes and initial open coding based on content. Next, open codes with similar meaning were combined. Finally, patterns within categories were analysed (e.g., barriers or facilitators for implementing and adhering to staff-provided PBS).

The results are displayed according to the predetermined codes, while also incorporating the findings from the open codes. For the purpose of this paper, quotes selected for the results section were translated from Dutch to English by the first author.

3 | Results

3.1 | Participants

In total, 13 frontline staff members from six residential group homes applied for participation in the study. Frontline staff worked in residential group homes with 5–14 people with intellectual disabilities and challenging behaviours. The number of frontline staff present at the group homes and providing daily care ranged from one to eight. A frontline staff member from the residential group home with the most applicants ($n=4$) was randomly drawn for exclusion, leading to a total number of participants of 12. The demographic characteristics of these frontline staff members are presented in Table 2. In the focus group, nine of the 12 frontline staff members continued to participate. Three frontline staff members were unable to attend due to scheduling problems.

Nine trainers applied for participation in the focus group, and all were included. Two trainers were the psychologists of teams of staff that participated in the trial; seven were external trainers. The demographic characteristics of the trainers are presented in Table 3.

3.2 | Comparison Current Sample and Trial Sample

No differences were found between frontline staff members of the current sample ($n=12$) and frontline staff members who received training in the trial ($n=129$) on age ($F(1, 129)=0.96$, $p=0.33$), sex ($F(1, 129)=3.04$, $p=0.08$), educational level ($F(1, 129)=0.32$, $p=0.58$), work experience in years ($F(1, 129)=0.29$, $p=0.59$), or the subscale Importance of the READI-SF ($F(1, 129)=1.26$, $p=0.29$).

3.3 | Results Semi-Structured Interviews With Frontline Staff Members

The results discussed in the current paragraph are grouped and presented according to the categories and determinants of the MIDI-ID.

3.3.1 | Critical Factors Regarding the PBS Training

In general, frontline staff members were positive about the PBS staff training. The training and the manual were perceived as clear, and frontline staff members found the theoretical background interesting, informative, and correct. All necessary materials were present during the training. Frontline staff members differed in their opinions about the difficulty level of the training. The majority thought the difficulty level was just right ($n=7$); some found it too difficult ($n=2$), while others considered it too easy ($n=3$). Only half of the frontline staff members had read the complete intervention manual.

> It was not the easiest training. It was no basic training, it was quite profound. But that gave room for good conversation. We experienced the training as useful.
>
> (Frontline staff member 5)

Frontline staff members perceived the training to be compatible with their care as usual. They expressed the belief that the training was effective in reducing challenging behaviours of adults with intellectual disabilities. Frontline staff perceived improvements in the behaviours of the people with intellectual disabilities for whom they cared

TABLE 2 | Demographic characteristics of included frontline staff members who participated in interviews and a focus group.

Characteristics	n (%)	Mean	Range
Age in years		40.1	25–57
Sex			
Male	7 (58.33)		
Female	5 (41.67)		
Work experience in years		12.1	2–30
Educational level			
Level 1: general secondary education	8 (66.67)		
Level 2: senior secondary vocational education	4 (33.33)		

TABLE 3 | Demographic characteristics of included trainers.

Characteristics	n (%)	Mean	Range
Age in years		42.8	34–61
Sex			
Male	0 (0)		
Female	9 (100)		
Educational background			
Applied behavioural therapist[a]	1 (11.11)		
Behavioural therapist[a]	6 (66.67)		
Psychologist	2 (22.22)		
Number of staff teams trained[b]		2.11	1–4

[a]As acknowledged by the Dutch Association for Behavioural Therapy and Cognitive Therapy (in Dutch Vereniging voor Gedragstherapie en Cognitieve Therapie).
[b]For the main trial.

> I remember we adjusted the schedule in the morning. We changed the start of the day. I saw great improvement in that. She [the resident, author] was much calmer and more relaxed. That's a good outcome.
>
> (Frontline staff member 10)

Although most frontline staff members ($n=8$) found the learned techniques applicable to the population they work with, some ($n=3$) questioned if the training would fit people with intellectual disabilities with multiple and more complex problems (e.g., comorbid speech disorders, movement disorders, autism spectrum disorders). One frontline staff member thought that the PBS staff training might be less suitable for people with severe or profound levels of intellectual disabilities.

3.3.2 | Critical Factors Regarding the Users

The majority of frontline staff members perceived that they learned how to 'consciously observe behaviour' and 'systematically look for solutions' ($n=8$). Moreover, some frontline staff members ($n=3$) felt that frontline staff-provided PBS contributed to more personal rest and more calmness at the residential group home.

> You consciously observe certain behaviour, and how to prevent that behaviour from happening. You observe with somewhat different glasses trying to reduce or improve behaviour.
>
> (Frontline staff member 1)

Some personal disadvantages that were mentioned were the loss of days due to the PBS staff training ($n=1$) and feelings of frustration due to unclear expectations from management on the utilisation and added value of PBS ($n=1$).

All frontline staff members expected that the people with intellectual disabilities and challenging behaviours under their care, as well as frontline staff members themselves, would benefit from the training. They ($n=6$) also had the impression that being trained in PBS resulted in clearer rules, more structure, and more calmness.

> The client I worked with has less stress because she is less restrained. Constantly sending her away caused lots of tension, and that was wearing. It is an advantage that has stopped.
>
> (Frontline staff member 9)

All frontline staff members considered the content of the training congruent with their profession. Regarding the cooperation of the people with intellectual disabilities they cared for, frontline staff members had different experiences. Some included the people in homework assignments ($n=4$), whereas others did not ($n=5$). They perceived observation assignments or the recording of interactions with the people as inconvenient, as the people with intellectual disabilities would not cooperate or would act differently when being filmed.

Seven frontline staff members experienced support from their co-workers to apply PBS techniques after the training, while five frontline staff members did not. They stated that their co-workers never understood the importance of PBS. The majority of frontline staff members ($n=8$) thought that their co-workers who participated in the training did not use the PBS techniques any more.

> It fades. I try to remind my co-workers. We would give each other compliments. Like, you do a good job. I also tried to compliment the clients when they were showing adaptive behaviour. Like, good job sitting here nice and cosy. Unfortunately, that declined.
>
> (Frontline staff member 6)

Regarding expectations that managers and involved psychologists might have about the utilisation of PBS, most frontline staff members felt that their manager and the psychologist involved expected them to still apply PBS techniques ($n=9$). Approximately half of the frontline staff members ($n=7$) stated that the PBS plans were still evaluated during team meetings.When asked about self-efficacy, the majority of frontline staff members stated that they currently could not apply the learned techniques without the manual ($n=8$). Half of the frontline staff members admitted that they had not read the complete manual. Regarding the fit between prior knowledge and the content of the innovation, most frontline staff members were positive. They perceived the trainin asg fitting to their educational level and they understood the rationale of frontline staff-provided PBS ($n=9$).

3.3.3 | Critical Factors Regarding the Organisation

When asked about formal ratification by management, most frontline staff members felt supported by their psychologists or direct manager ($n=11$). However, all frontline staff members reported that PBS was not formally incorporated in the policy of their organisations. One frontline staff member stated that her manager communicated from the start of the training that they would not proceed with PBS.

Regarding staff turnover, all frontline staff members stated that new frontline staff members were not informed about PBS. The majority of frontline staff members ($n=9$) believed that teams with high staff turnover were less likely to apply PBS. When asked about current staff capacity, the majority of frontline staff members were satisfied, yet two frontline staff members indicated they could not apply PBS techniques due to understaffing. Two other frontline staff members perceived that utilisation of PBS techniques was not determined by the number of frontline staff, but rather by the consistency of frontline staff.

Financial resources were not a problem according to most of the frontline staff members ($n=9$). Also, the majority of frontline staff members ($n=10$) were able to reach a good balance between daily workload and putting PBS techniques into practice, although two frontline staff members could not apply the PBS techniques due to their high workload and limited time. These two frontline staff members also pointed out that more financial resources were probably necessary to facilitate more time.

Regarding material resources and facilities, all frontline staff members indicated that the training manuals were still present at the residential group homes. Additionally, they felt as if they could consult their psychologists if they had questions about PBS techniques.

The majority of frontline staff members pointed out that education or training in other methods and techniques ($n = 11$), staff turnover ($n = 4$), moving of the residential group home ($n = 2$) and budget cuts ($n = 5$) negatively affected utilisation of PBS. Finally, all frontline staff members would have liked to receive direct feedback from their management or the research team regarding the effect of the training.

3.3.4 | Critical Factors Regarding Socio-Political Environment

According to the frontline staff members, the PBS staff training was in line with the Convention on the Rights of Persons with Disabilities (United Nations 2006) and with the recent Dutch law regarding restraining measures (in Dutch 'Wet zorg en dwang'; Ministry of Health, Welfare and Sport 2020) which dictates to use restraining measures as a last resort only. Frontline staff members ($n = 4$) reported that the use of functional analyses helped to reduce challenging behaviours of the people with intellectual disabilities, which in turn decreased the use of restraint.

3.4 | Results Frontline Staff Focus Group

3.4.1 | Perceived Requirements for Implementation

According to frontline staff, the most important requirement for the implementation of frontline staff-provided PBS was that it should be embedded in the whole organisation. The highest layer of management of their organisation should propagate PBS. PBS should recur in the language, be mentioned on the website, and be included in the objectives, plans, and actions of the annual policy plans of the organisation. According to the frontline staff members, organisational embedding also means that management has knowledge of the theoretical background and rationale of PBS and that they are able to explain these to the teams.

> Management should have substantive knowledge of PBS so they can support me and my colleagues.
>
> (Frontline staff member 9)

The focus group with frontline staff also stated that the complete team of frontline staff should be unanimous, motivated, and ready to participate in the PBS staff training. Thus, successful preparation and motivation of teams by management to implement frontline staff-provided PBS were perceived as key. To enhance willingness to participate in PBS, frontline staff put forward that it is essential for them that management clearly articulates the rationale behind PBS, providing the opportunity to ask questions and express any hesitations. According to the frontline staff members, psychologists play a crucial role in facilitating this process by engaging with staff concerns directly and supporting their motivation. Furthermore, they emphasised the importance of encouraging and supporting each other in adopting and implementing PBS in their residential group home.

> If a team is aware of the rationale and the 'why', then it sticks more.
>
> (Frontline staff member 1)

Finally, according to frontline staff, effective implementation of staff-provided PBS necessitates substantial support from management in terms of time, space, and financial resources. Specifically, sufficient time is deemed essential for the development and discussion of PBS support plans and for additional training sessions. Moreover, providing office space away from the residential group home is perceived as crucial to ensure open discussions about PBS support plans. Financial investment was also mentioned to be required to meet these logistical needs. According to frontline staff, by meeting these requirements, management not only demonstrates its commitment to PBS but also signals a long-term investment, thereby enhancing staff motivation and engagement.

3.4.2 | Perceived Requirements for Adherence

Regarding adherence to frontline staff-provided PBS, frontline staff perceived an important role for the psychologists: they should support the team, be present during the PBS training meetings, set the right example, and help with formulating the PBS plans. Moreover, psychologists should be regularly present in the residential group home to support the adherence of frontline staff to the PBS plans.

> If I struggle, it would be nice if the psychologist could give an example.
>
> (Frontline staff member 12)

Another important factor for adherence mentioned by the frontline staff members was the documentation and visibility of the PBS support plans. These should have a prominent place in the electronic patient files, combined with physical reminders (such as binders) within the residential group home. These measures were perceived to simplify the documentation and review of support procedures, enhance the visibility of PBS, and consequently serve as constant reminders to adhere to the support plans. Moreover, frontline staff members thought PBS support plans should be easily accessible and clearly written so that new staff members know what is expected from them and are encouraged to apply PBS practices.

> If we want to keep working with the PBS plans, new staff should be properly informed.
>
> (Frontline staff member 4)

Finally, recurring evaluations of the PBS support plans during team meetings were considered necessary by frontline staff. They felt that frontline staff and psychologists should actively collaborate during such meetings. Frontline staff could offer insights based on their daily interactions with and observations of the people they care for and provide feedback on the effects of the PBS support plans. Psychologists could add their theoretical knowledge of PBS techniques to the meetings and, if necessary, refine and improve the support plans. Frontline staff members as well as the psychologists should feel responsible for these meetings.

3.5 | Results Trainers Focus Group

3.5.1 | Requirements for Implementation

Time was also an important requirement for implementation according to all of the trainers. Time needs to be available to conduct the training, and there should perhaps be more time in between training sessions (i.e., 3 weeks instead of 2 weeks). Moreover, during the training sessions, time should be spent on the training rationale, not on other matters such as daily matters.

> Make sure that the time you have is spent on PBS and not on daily hassles.
>
> (Trainer 8)

Motivation of frontline staff members was also considered an important requirement for implementation. The trainers stated that frontline staff members should be willing and motivated to participate in the PBS staff training. Limited frontline staff motivation or even negative assumptions about PBS not only negatively influence the frontline staff member themselves, but also their team members. The trainers believed that providing information about the rationale of PBS prior to the staff training might help to overcome these barriers.

The trainers also pointed out that more than one frontline staff member should be specifically assigned to and involved with the individual with intellectual disabilities who is targeted in the PBS staff training. This way, frontline staff members can better cooperate on PBS plans. The trainers believed that this would reduce the workload for the frontline staff members and better assure the continuity in applying the PBS plans.

The psychologists of the group homes play an important part in the PBS frontline staff training according to the trainers. They should be present at the training sessions and support the trainers. The trainers found it preferable that the psychologist has some background or knowledge of PBS and behavioural principles.

Finally, the trainers suggested that the organisation should make clear choices on what methods or interventions should be prioritised organisation-wide. Combining multiple methods or interventions is possible, but too many would be undesirable. When clear choices are made, more time is available for training and applying PBS.

> You're part of a complete organisation. You work with a team of staff, with a manager, with supporting staff such as medical service. You're part of a web. The organisation should choose which way everyone goes.
>
> (Trainer 4)

3.5.2 | Perceived Requirements for Adherence

Regarding the adherence to frontline staff-provided PBS, trainers mentioned time as the most important requirement. Time should be available prior to the PBS frontline staff training for the trainers or management to explain the rationale for frontline staff-provided PBS. Moreover, time should be reserved during staff meetings for the evaluation of the PBS plans.

The trainers also suggested shorter and summarised PBS plans. Shorter plans might be easier for frontline staff to execute and, as a result, the PBS principles might become more integrated in daily practice when using shorter plans.

Additionally, according to the trainers, the PBS plans should be well documented in the patient files or daily reports and easily accessible. Clear and specific guidelines for behavioural management were considered the greatest value of PBS.

Finally, the trainers stated the importance of ongoing education, as in a refresher course in PBS for the frontline staff members, perhaps provided online. Moreover, psychologists should receive additional training in behavioural principles and PBS, and all new frontline staff members should be trained in PBS.

> It takes a lot to properly educate and coach the staff members. For proper embedding, a PBS staff training of a couple of weeks is not enough.
>
> (Trainer 8)

4 | Discussion

In this mixed-methods study, we aimed to identify key factors in implementing and adhering to frontline staff-provided PBS in the care for adults with intellectual disabilities and challenging behaviours. We used interviews to explore frontline staff's experiences with a PBS training, as well as their perspectives on factors influencing the implementation of and adherence to PBS. Focus groups with frontline staff members and PBS trainers were conducted to further broaden our insight into factors that may promote the implementation and adherence of staff-provided PBS.

Regarding frontline staff's perspectives on the PBS training, our results show that frontline staff members were positive about the PBS training and observed improvements in the challenging behaviours of the people they cared for. These findings are in accordance with studies demonstrating the effect of PBS on the reduction of challenging behaviours of people with intellectual disabilities (Bruinsma et al. 2024; MacDonald and McGill 2013; Grey and McClean 2007) Also, after PBS training, frontline staff members experienced more calmness within themselves, and observed similar changes among their colleagues and the people they cared for. It could be that frontline staff members have experienced enhanced perceived control over challenging behaviours after the PBS training, which may have resulted in calmer behaviour of themselves, which, in turn, may have contributed to increased calmness and relaxation in the people they supported. This aligns with previous research, that demonstrated a shift in staff's emotional responses to challenging behaviours following PBS training (Lowe et al. 2007; MacDonald and McGill 2013). Moreover, increased staff perception of understanding of and control over how their own behaviours influence challenging behaviours is associated with reductions in such behaviours among people with intellectual disabilities (Griffith and Hastings 2014; Hastings et al. 2018). However, further research is needed to obtain a more thorough understanding of whether changes in frontline staff's emotional responses and their understanding of challenging behaviours mediate the effects of PBS on people with intellectual disabilities.

Despite these positive experiences, some frontline staff regarded the PBS training to be challenging. Furthermore, although they indicated relying on the training manual to apply PBS, only half of them actually read the complete manual. Previous studies have highlighted the difficulties faced by frontline staff members in comprehending the PBS principles and the subsequent documentation (Adkins et al. 2002; Singh et al. 2009; Wardale et al. 2018). Realising the objectives of PBS (i.e., improving quality of life and reducing challenging behaviours) is difficult if frontline staff members fail to grasp the underlying principles of the intervention. Other complicating factors mentioned by frontline staff were limited time, limited motivation of staff, low peer support, and high staff turnover. This aligns with a recent study by Brady et al. (2024), who found that while frontline staff expressed generally positive views about PBS, they encountered significant barriers in practice. These findings show that the utilisation of frontline staff-provided PBS remains challenging.

Frontline staff members as well as trainers considered organisation-wide embedding of PBS a crucial prerequisite for successful implementation of PBS. This finding is in accordance with previous research that highlighted the importance of a multidimensional approach for PBS (Lowe et al. 2007; McKenzie et al. 2020), and a recent definition of PBS in which the multi-tiered aspect of PBS is adopted (Gore et al. 2022). Brady et al. (2024) similarly highlighted organisational constraints as a barrier to PBS implementation, noting that institutional hierarchies and power divides between frontline staff and clinicians hindered successful implementation. Hayward et al. (2021) further emphasise the need to incorporate PBS into the organisation's values and framework to ensure successful implementation of frontline staff-provided PBS. According to frontline staff members in the current study, organisation-wide embedding should also ensure that management understands the PBS rationale or receives necessary PBS training. This approach resonates with the concept of practice leadership, defined as “an individual (usually a frontline manager) who develops, encourages and supports their staff team to put into practice the vision of the organisation” (Beadle-Brown et al. 2015, 1082). Research indicates that effective practice leadership is associated with enhanced staff skills (Beadle-Brown et al. 2015).

The acceptability of PBS was another important factor for facilitating implementation according to both frontline staff members and trainers. Sufficient time should be taken to inform and motivate frontline staff members to participate in PBS training and to adhere to PBS practices. As negative prejudice among frontline staff against behavioural interventions such as PBS has been reported (Ager and O'May 2001; Gormley et al. 2019), it is important for the implementation of frontline staff-provided PBS that management takes time to clarify the rationale of PBS and provides room to discuss and solve possible adversity of frontline staff members. In the Netherlands, PBS is hardly used in disability services, and this unfamiliarity may have negatively affected the implementation of and adherence to PBS. Some frontline staff members in the current study expressed they observed a lack of interest in utilising PBS among their colleagues. Consequently, our results may not be applicable for or generalisable to other countries more familiar with PBS, such as the United Kingdom.

Regarding adherence to PBS, frontline staff and trainers emphasised that PBS plans should be easily accessible and understandable to all staff, particularly for new employees. Additionally, they addressed the possibility of incorporating PBS plans into the current digital documentation systems to facilitate easy recording. Also, previous studies stated the importance of clear documentation and the need to adapt existing monitoring systems to behavioural interventions (Beadle-Brown et al. 2012; Francke et al. 2008; MacDonald and McGill 2013). Another factor for adherence mentioned by frontline staff and trainers was time to regularly evaluate the PBS plans. This is in line with previous research (McKenzie et al. 2020). According to both focus groups, the psychologist involved should play an important role in these evaluations. Earlier research also showed that psychologists' support and leadership are crucial in managing challenging behaviour (Olivier-Pijpers et al. 2019; Stenfert Kroese and Smith 2018) and their educational background currently better fits the PBS rationale and documentation of PBS plans (Wardale et al. 2018). However, the trainers noted the need for better education of the psychologists on PBS to provide sufficient support for the frontline staff members. Possibly, a train-the-trainers model could meet that need (MacDonald and McGill 2013), as research showed no differences in effect between first- or second-generation PBS training (Lavigna and Willis 2012). Hence, training psychologists could result in better embedding PBS within the organisation, increasing PBS practice by frontline staff and training new frontline staff members through a train-the-trainer model. This would tackle another problem that frontline staff members addressed in the interviews: new frontline staff members were not trained or instructed to apply PBS. Frontline staff turnover is a well-known problem in the care of people with intellectual disabilities and challenging behaviours (Mulhall et al. 2021), which may complicate adherence to PBS (Bosco et al. 2019).

Although the current study provided valuable insight into frontline staff and trainers' perspectives on factors related to the implementation of and adherence to PBS, some limitations should be mentioned. First, only frontline staff members and trainers who participated in the training were included in the study. Broader perspectives could have been obtained by including stakeholders such as people with intellectual disabilities and challenging behaviours themselves, legal guardians, or managers. A systematic approach that combines and synthesises knowledge and experiences from different stakeholders is essential for improving the care of people with intellectual disabilities (Man and Kangas 2020). Second, we only included a small number of frontline staff members in our study, compared to the larger number of frontline staff that were trained for the trial. While this sample size is sufficient for (semi-structured) interviews (Hennink and Kaiser 2022; Sandelowski 1995), more than two focus groups may be necessary to achieve saturation (Hennink and Kaiser 2022). Moreover, the number of participants may be perceived as limited for a mixed-methods approach. Furthermore, frontline staff members from only six residential group homes participated in the current study, while teams of frontline staff from fourteen residential group homes were eligible. Although the frontline staff members in the current study did not significantly differ in age, sex, educational level, work experience, or willingness to apply new interventions from those participating in the trial, it is possible that frontline staff members who were more positive about the training were also more willing to participate in the current study (Campbell 2007). As a result, the findings and their generalisability should be interpreted with caution.

To conclude, this study highlights factors associated with frontline staff and trainers perspectives on the implementation of and adherence to frontline staff-provided PBS. While frontline staff generally held positive views about the PBS training, implementing and adhering to PBS appeared to be challenging in many respects. Key factors in promoting intervention implementation and adherence were to embed PBS within the entire organisation, actively support frontline staff members by psychologists, promote the acceptability of the intervention, motivate frontline staff members before the training, provide clear documentation, dedicate time for evaluation, and ensure ongoing education and support for frontline staff members. Awareness of these factors can help clinical practice with the uptake of frontline staff-provided PBS in the care for people with intellectual disabilities and challenging behaviours.

Author Contributions

E.B. designed and directed the project. A.A.d.B. and B.J.v.d.H. contributed to the data collection. All authors aided in interpreting the results and provided critical feedback and helped shape the research, analysis, and manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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